ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 6

Information and Communication Technology and Society

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

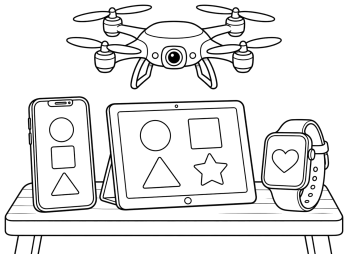
Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain how ICT is used in films, music and games.
- [ ] Name the issues associated with the use of ICT.
- [ ] Explain data theft, unauthorized access, intellectual property and fraud.
- [ ] Avoid plagiarism by citing, quoting and referencing.
- [ ] Protect computers with physical and logical security.
- [ ] Name harmful software (malware) and keep a computer safe from it.
- [ ] Explain ergonomics and four health issues of computer use.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

3D	Holographic image	Simulation	Digital audio
Data theft	PII	Unauthorized access	Patent
Fraud	Plagiarism	Citing	Referencing
UPS	Surge protector	Firewall	Backup
Malware	Trojan horse	Phishing	Cyber security
Ergonomics	Posture	RSI	CVS

"Use ICT to make life better - for yourself and for everyone around you."

- THE KING OF ICT

WRITING SPACE 1 - The four parts of this unit

write what your teacher tells you

CHAPTER 1

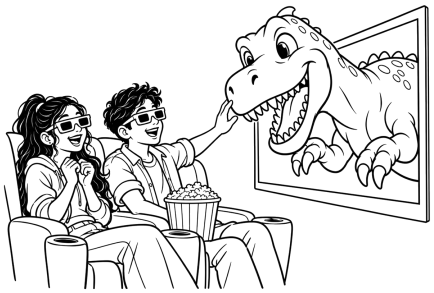
ICT in Entertainment

A DINOSAUR IN THE CINEMA!

"We watched a film. A dinosaur jumped out at us!"

"That was a 3D film. ICT makes films look real."

ICT has changed films, music and games. Meet six modern facilities of entertainment.



6.1 ICT in films

To relax from everyday stress, most people prefer watching a film. Unlike in the past, film producers now make films of **high quality** with the use of ICT:

	Technology	What it is
	3D - three dimensional	Makes life look real on films (films were two dimensional once). The viewer needs **3D spectacles**.
	Holographic image processing technology	A scene shot in one place is recorded on camera and displayed **elsewhere**; mostly in **horror** scenes.
	Cartoon films	Popular among adults and the young: the **3D feature** and new technologies in the software.

WRITING SPACE 2 - 3D films and holographic images write what your teacher tells you

6.2 ICT in music and games

Computers are also used for modern music and its editing, for digital games and for simulation games:

	Technology	What it is
	Digital audio materials	Songs recorded, **rhythm** created and **mixing** done with software; stored on **CDs**, played on **home theatre systems**.
	Digital games	Played on **computers** or **cell phones**. Games in **3D** and **4D** are the most popular today.
	Simulation games	An everyday activity in a **natural setting created artificially**; used for training: sports activities, investigations, planning, military training, mercantile activities, miming.

WRITING SPACE 3 - Games and simulation games write what your teacher tells you

A Match them

6 marks

Write the correct letter in the box.

A - Technology		B - What it is
1. 3D technology		**A.** Recording songs, creating rhythm and mixing.
2. Holographic image processing		**B.** A natural setting created artificially, for training.
3. Cartoon films		**C.** Makes films look real; viewers need spectacles.
4. Digital audio materials		**D.** Popular with adults and the young: the 3D feature.
5. Digital games		**E.** Played on computers or cell phones; 3D and 4D.
6. Simulation games		**F.** A scene recorded in one place is displayed elsewhere.

B Fill in the blanks

8 marks

Use the words in the box. Two words are extra.

WORD BANK

film	3D	spectacles	horror	mixing
discs	theatre	4D	2D	cartoons

1. To relax from everyday stress, most people prefer watching a ..... .
2. With ..... technology, producers make life look real on films.
3. To view a 3D film, the viewer needs 3D ..... .
4. Holographic images are mostly used in scenes depicting ..... .
5. Recording songs, creating rhythm and ..... are done with software.
6. Recorded songs can be stored in compact ..... (CDs).
7. We can listen to the songs on home ..... systems.
8. Digital games in both 3D and ..... are the most popular today.

"Enjoy films, music and games - but never forget your studies."

- THE KING OF ICT

C True or false

6 marks

Write T (true) or F (false) in the box.

1.		Film producers today make films of high quality with the use of ICT.
2.		You do not need any spectacles to view a 3D film.
3.		Cartoon films are popular among both adults and the young.
4.		Digital games can be played only on computers, not on cell phones.
5.		Simulation games are used for military training.
6.		Holographic image processing is mostly used with comedy scenes.

D Choose the correct answer

5 marks

Underline your answer.

1. What does 3D stand for?
   (a) three digital (b) three dimensional
   (c) third disc (d) three devices

2. Which technology records a scene in one place and displays the image elsewhere?
   (a) digital audio (b) simulation games
   (c) holographic image processing (d) cartoon films

3. Songs recorded with modern software can easily be stored in
   (a) compact discs (b) printed books (c) photo albums (d) letters

4. Simulation games are based on an everyday activity in
   (a) a cartoon film (b) a compact disc
   (c) a home theatre (d) an artificial natural setting

5. Which of these is NOT a use of simulation games given in the textbook?
   (a) military training (b) storing songs (c) planning (d) sports training

"A good film relaxes the mind. A good book feeds it."

- THE KING OF ICT

CHAPTER 2

Ethical and Legal Issues

? IS IT REALLY YOUR ESSAY?

"I will hand in this essay from a website as mine!"

"Don't! That is plagiarism. Name the owner."

ICT also brings problems. Meet the legal issues and the issues related to values.



6.3 Problems of using ICT

ICT can be mankind's closest friend, with many facilities that make life comfortable. Yet, with the use of ICT, mankind faces many unforeseen problems:

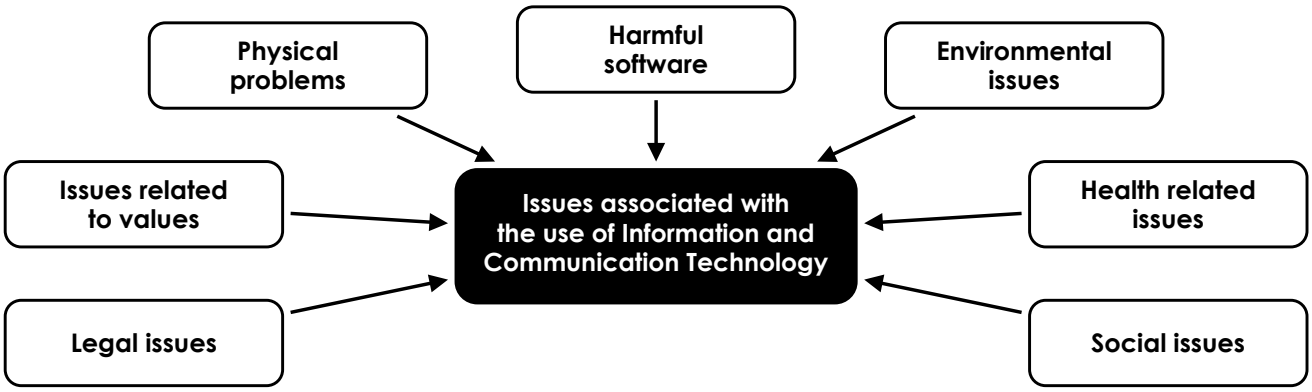


Figure 6.8 - issues associated with the use of ICT

WRITING SPACE 4 - Problems of using ICT write what your teacher tells you

6.4 Legal issues

Legal issue	What it is	Example / note
Data theft	Personally Identifiable Information (**PII**) is stolen from a computer or another device.	Stealing or keeping flash drives, digital cameras or mobile phones with personal names, telephone numbers and photos, without permission or legal rights.
Unauthorized access	Accessing a computer system unlawfully by stealing or manipulating the **user name** and **password**.	An organization's data can easily be stolen in this way.
Intellectual property rights	Intellectual property relates to **creations of the mind**: an innovation or a new product. It belongs to the **first** person or organization to develop and release it.	Using or selling it without the owner's knowledge is illegal. A **patent** - a license issued by the **state** for innovations - protects it.
Fraud	Copying or impersonating personal documents (bank account numbers, signatures) on a CD or over the Internet.	Copying literary works for personal benefit; cheating people with business transactions; credit card fraud.

"Honesty is the best software. Never copy another person's work."

- THE KING OF ICT

6.5 Issues related to values: plagiarism

Plagiarism is the act of stealing another's creative work and displaying it as one's own. It is very common today, with data and information stolen from the Internet. Use information from the Internet in **better accepted ways**, without harming the reputation of its owner:

Way	What you do
Citing	Mention the rightful owner and his information.
Quoting	Use inverted commas ("...") to mark a selected or borrowed section.
Referencing	List the resources from which the information was collected - usually at the end of an essay or article.

AN EXAMPLE

The textbook says that plagiarism is **"the act of stealing another's creative work and displaying it as one's own"** *[quoting]* (Grade 11 ICT textbook, page 172) *[citing]*.

References: Information and Communication Technology, Grade 11 textbook, page 172. *[referencing - at the end of the essay]*

WRITING SPACE 5 - Plagiarism

write what your teacher tells you

A Which legal issue?

8 marks

Which legal issue is it? Tick one box in each row.

It is ...	Data theft	Unauthorized access	Intellectual property	Fraud
Stealing a flash drive with customers' PII.	☐	☐	☐	☐
Logging in with a stolen user name and password.	☐	☐	☐	☐
Selling another person's new product as yours.	☐	☐	☐	☐
Using another's signature and account number.	☐	☐	☐	☐
Keeping a phone with other people's photos.	☐	☐	☐	☐
Manipulating a manager's password to read data.	☐	☐	☐	☐
Using an invention without the owner's knowledge.	☐	☐	☐	☐
Cheating people with a business transaction.	☐	☐	☐	☐

"Give credit to the owner. Your own words are worth more."

- THE KING OF ICT

B Citing, quoting or referencing?

6 marks

Tick one box in each row.

Which way is it?	Citing	Quoting	Referencing
Nimal names the writer of the article he used.	☐	☐	☐
Kasuni puts "..." around a sentence she borrowed.	☐	☐	☐
Dilani lists her websites at the end of her essay.	☐	☐	☐
Saman names the rightful owner under a photo.	☐	☐	☐
A borrowed part of a speech is put in "...".	☐	☐	☐
A list of the books used ends an article.	☐	☐	☐

C Choose the correct answer

5 marks

Underline your answer.

1. PII stands for
   (a) Personally Identifiable Information (b) Private Internet Information
   (c) Personal Identity Index (d) Public Information Instruction

2. A patent is a license issued by
   (a) a bank (b) a school (c) the state (d) a website

3. Which of these is an issue related to values?
   (a) data theft (b) plagiarism (c) fraud (d) hacking

4. Listing the resources from which information was collected is
   (a) citing (b) quoting (c) plagiarism (d) referencing

5. Which of these is NOT one of the issues in Figure 6.8?
   (a) legal issues (b) social issues (c) fast e-mail (d) health issues

D Answer these

4 marks

Write full sentences.

1. What are intellectual property rights? (2 marks)

2. Write two better accepted ways of using information collected from the Internet. (2 marks)

CHAPTER 3

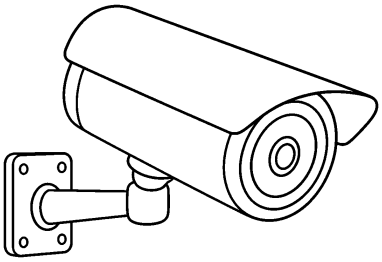
Protecting ICT Infrastructure

? WHO WAS IN OUR LAB?

"Someone took a computer from our lab at night!"

"Locked doors and CCTV cameras protect a lab."

Careless use of computers brings physical and logical issues. Learn to protect computers, data and networks.



6.6 Physical security

Careless use of computers brings **physical** and **logical** issues: sudden power failures damage devices, there may be computer thieves, malware from the Internet can break down a whole network, and hacking and environmental factors damage computers. **Physical security** protects the computers:

Physical security	What it does
UPS (Uninterrupted Power Supply)	When the electricity fails, the **battery** in the UPS supplies power, so the computer, the network and the documents face no harm.
Hardware firewall	Now built into **broadband routers**. Essential for networks connected to the Internet: it filters access and protects a network from **unauthorized access**.
Controlled access through locked doors	Closed and locked doors prevent or limit the unlawful entry of people into a computer laboratory.
CCTV	Cameras inside a computer laboratory monitor movement from outside, or record the day's proceedings as a video clip for viewing later.
Surge protector	The local voltage supply is **240 V**. The surge protector acts when the supply exceeds this limit and protects electronic devices from destruction.
Environmental factors	Keep a laboratory or a computer in a safe environment, free from **dirt**, **dust** and **moisture**.

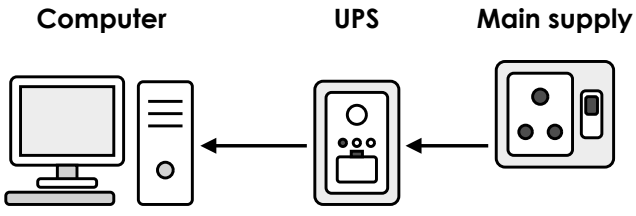


Figure 6.11 - Uninterrupted Power Supply (UPS)

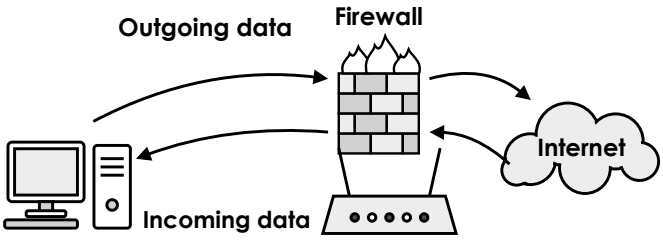


Figure 6.12 - a hardware firewall

"Protect your data today, and you will not cry tomorrow."

- THE KING OF ICT

WRITING SPACE 6 - Physical security

write what your teacher tells you

6.7 Logical security

Logical security	What it does
Passwords	Secure the data and stop unauthorized entry or use. Mix **letters**, **symbols** and **numbers** to make a password stronger.
Software firewall	Part of the **operating system**; it must be active at all times. It checks all incoming data for authenticity and stops suspicious data.
Backups	Keep a copy of essential data on an external hard disk, CD, DVD, flash drive or memory chip - securely, in **another physical location**.

6.8 Malicious software (malware)

The use of a computer involves various threats. **Malicious software** (**malware**) or malicious codes cause serious harm to computers and networks: it affects the efficiency of the computer (improper function, restarting again and again); destroys software or makes it malfunction; stops other software from being installed; weakens the hardware; sabotages computer networks; steals and destroys data; and fills the hard disk with unnecessary files.

6.9 Harmful software and the damage it does

Harmful software	The damage it does
Computer virus	Enters through a program (networks, USB flash drives, memory chips, e-mail) as an **executable file** and spreads rapidly within the computer.
Computer worm	Acts like a virus, but **spreads alone** through e-mail attachments, false websites and instant messages (social engineering).
Trojan horse (spyware)	Looks harmless and enters with e-mail attachments, without the user's knowledge; opens windows, deletes documents, steals data. It **does not spread by itself**.
Adware	Displays unnecessary notices (advertisements) and collects commercial information. Not as harmful as viruses, but it disturbs the user.
Bots	From "robots". Work on their own with other networks and collect personal information through Internet messages.
Browser hijacker	Misdirects a user to a **different website** to collect information on trade, commerce and advertisements.
Phishing	Deceives users with an **e-mail** and a link (as if from a popular organization or a friend) to steal account details and cash.
Spam	**Unauthorized e-mail**, mostly advertisements. It fills the mail box and may collect e-mail addresses for frauds.

WRITING SPACE 7 - Virus, worm and Trojan horse write what your teacher tells you

6.10 Safeguarding a computer and a network

Prevention is better than cure. Always:

- Install a virus guard, update it and keep it always active.
- Check USB memory with anti virus software.
- Always install authorized software.
- Use a user account, not an administrator account.

If the computer is connected to the Internet:

- Access secure websites; check the URL.
- Download only from secure websites, after a virus check.
- Never click suspicious links or open suspicious e-mails.
- Do not enter personal information without checking on security.
- Use a firewall, virus guards and e-mail filters.

Some secure, popular virus guards: Avira Antivirus, Avast Antivirus, AVG Antivirus, K7 Antivirus, Digital Defender Antivirus, Norman Antivirus, Kaspersky Antivirus, Panda Cloud Antivirus, Microsoft Security Essentials, Norton Antivirus, BitDefender Antivirus, McAfee Antivirus.

DO IT - PRACTICAL 1 (TEXTBOOK ACTIVITY, PAGE 178)

Virus guards on the Internet. Work in pairs in the computer laboratory, with your teacher.

1	Search the Internet for virus guards that are not in the list of 6.10.
2	Use only secure websites. Do not download or install anything.
3	For each virus guard, find the name of its manufacturer and its date of manufacture.
4	Tabulate them in the table below. Compare your list with the class.

Virus guard	Manufacturer	Date of manufacture

"Back up today. Tomorrow may be too late."

- THE KING OF ICT

6.11 Cyber security in Sri Lanka

Almost every person and organization in Sri Lanka is connected to the Internet, and unauthorized access into social networks and financial websites is seen all over the world. So responsible agencies must secure Internet access: this is **cyber security** (also for personal networks and operating systems).

- **ICTA** (Information and Communication Technology Agency) is responsible for the **Institution for Information Security of Sri Lanka**: it serves citizens, business establishments and state institutions.
- ICTA, the **Sri Lanka Standards Bureau** and the **Sri Lanka Emergency Computer Services** certify **Information Security Management Systems (ISMS)**.

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITY, PAGE 179)

Computer related services in Sri Lanka. Do this with your teacher.

1	Open the three websites of the textbook (in the table below) with your teacher.
2	Check the URL of each website before you use it.
3	Make a list of the computer related services you can obtain from each website.
4	Do not enter any personal information on the websites.

Website	Computer related services I can obtain
http://www.gov.lk/web/	
http://www.engage.icta.lk	
https://www.techcert.lk/si/	

A Physical or logical?

8 marks

Is it physical or logical security? Tick one box.

Security measure	Physical security	Logical security
A UPS	☐	☐
A password	☐	☐
A CCTV camera	☐	☐
A software firewall	☐	☐
A surge protector	☐	☐
A backup on an external hard disk	☐	☐
Closed and locked doors	☐	☐
A hardware firewall in a broadband router	☐	☐

"A strong password is a strong lock. Never share the key."

- THE KING OF ICT

B Match the harmful software

8 marks

Write the correct letter in the box.

A - Harmful software		B - What it does
1. Computer virus		**A.** Unauthorized e-mail, mostly advertisements.
2. Computer worm		**B.** Looks harmless; does not spread by itself.
3. Trojan horse		**C.** Misdirects a user to a different website.
4. Adware		**D.** An executable file that spreads within the computer.
5. Bots		**E.** Displays unnecessary notices on the screen.
6. Browser hijacker		**F.** An e-mail and a link that steal bank account details.
7. Phishing		**G.** Spreads alone through e-mail attachments and messages.
8. Spam		**H.** Works on its own; collects personal information.

C True or false

6 marks

Write T (true) or F (false) in the box.

1. ☐ A firewall protects a network from unauthorized access.

2. ☐ A software firewall needs to be active only once a week.

3. ☐ A Trojan horse spreads by itself, like a computer virus.

4. ☐ Adware displays unnecessary notices on the computer screen.

5. ☐ Phishing uses e-mail to collect information about bank accounts.

6. ☐ A backup is safest when it is kept right next to the computer.

D Choose the correct answer

5 marks

Underline your answer.

1. Which device protects electronic devices when the power supply exceeds 240 V?
   (a) a UPS (b) a router
   (c) a surge protector (d) a CCTV camera

2. Which of these is a logical security measure?
   (a) a locked door (b) a password (c) a CCTV camera (d) a UPS

3. Which harmful software misdirects a user to a different website?
   (a) browser hijacker (b) computer worm
   (c) spam (d) adware

4. Which of these does NOT safeguard a computer from harmful software?
   (a) a virus guard (b) a user account
   (c) authorized software (d) suspicious links

5. Who is responsible for the Institution for Information Security of Sri Lanka?
   (a) a bank (b) ICTA (c) a school (d) a newspaper

E Word search

10 marks

Find the 10 words about harmful software and protection. Tick each one.

P	H	A	V	I	R	U	S	W	L	L	E
H	B	D	Y	S	D	D	B	Y	L	R	P
I	C	W	Y	W	P	A	T	A	E	W	C
S	F	A	W	O	Y	Y	W	I	Y	P	R
H	S	R	A	R	Y	E	W	E	G	P	R
I	T	E	P	M	R	P	T	A	I	R	U
N	O	C	K	I	M	K	M	B	R	C	S
G	B	I	F	T	R	O	J	A	N	E	I
I	R	O	P	M	B	C	H	W	Y	H	T
B	H	I	J	A	C	K	E	R	F	L	M
B	B	G	O	L	C	F	M	A	P	S	D
N	T	E	P	R	Y	C	E	C	M	S	A

FIND THESE WORDS

- ☐ VIRUS
- ☐ WORM
- ☐ TROJAN
- ☐ ADWARE
- ☐ SPYWARE
- ☐ BOTS
- ☐ HIJACKER
- ☐ PHISHING
- ☐ SPAM
- ☐ FIREWALL

CHAPTER 4

Health Issues and Ergonomics

? MY EYES HURT!

"I played games all evening. My eyes are red!"

"That sounds like Computer Vision Syndrome!"

Long hours at a computer bring health issues. Meet ergonomics and four health issues.



6.12 Ergonomics

Ergonomics is a word made of two words of the Greek language: **"ergon"** is job or action, and **"nomos"** is law. The simple meaning of the word is:

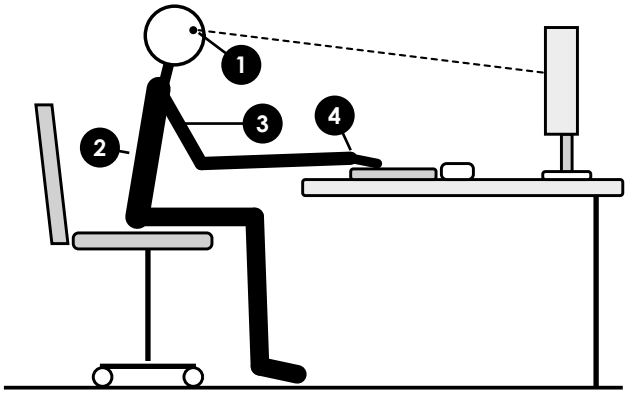
i ERGONOMICS

A job must be created to the comfort of the worker, and the worker must not be forced to adjust himself or herself to the work. If not, the worker will suffer various tensions and illnesses.

Technology and mankind are now inseparable, and more people use technology every day, so technology related health issues keep increasing. Issues arise from the continuous use of a computer for **over four hours at a stretch**.

WRITING SPACE 8 - Ergonomics write what your teacher tells you

6.13 Four health issues



1. **Computer Vision Syndrome (CVS)**
2. **Musculoskeletal problems**
3. **Repetitive Stress Injury (RSI)**
4. **Carpal Tunnel Syndrome (CTS)**

Health issue	What it is	The cause
Musculoskeletal problems	Pain in different muscles and bones of the body.	Non-stop use of a computer with the **wrong posture**.
RSI - Repetitive Stress Injury	Pain from the **shoulder to the fingers**; swelling and hardness; it is hard to move the mouse.	**Incorrect posture**.
CTS - Carpal Tunnel Syndrome	Numbness and pain in the **fingers**, from pressure on the wrist.	Incorrect use of the **keyboard** and the **mouse**, or placing them in incorrect positions.
CVS - Computer Vision Syndrome	Irritation of the **eyes**: dry eyes, redness, tearing, blurred vision, pain in the head, neck or back.	Sticking to the computer continuously for **6 to 7 hours**.

AVOID THESE PROBLEMS

Sit in the **correct posture**, place the keyboard and the mouse in the **correct positions**, and do not use a computer non-stop for hours at a stretch.

WRITING SPACE 9 - Four health issues write what your teacher tells you

A Which health issue?

8 marks

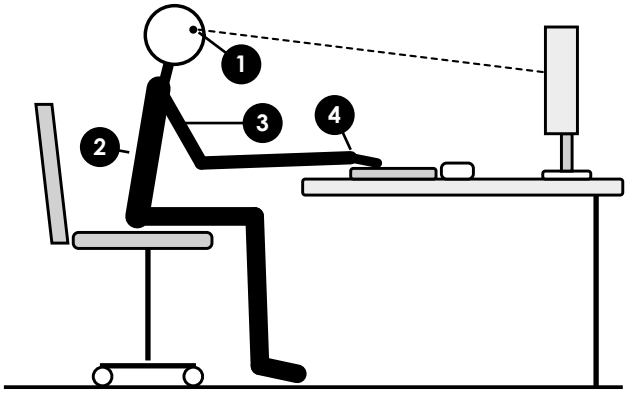
Which health issue is it? Tick one box in each row.

It is ...	Musculo-skeletal	RSI	CTS	CVS
Pain in muscles and bones from a wrong posture.	☐	☐	☐	☐
Pain extending from the shoulder to the fingers.	☐	☐	☐	☐
Numbness in the fingers from pressure on the wrist.	☐	☐	☐	☐
Dry eyes, redness in the eyes and blurred vision.	☐	☐	☐	☐
It is hard to move the mouse because of this pain.	☐	☐	☐	☐
The keyboard and mouse are in incorrect positions.	☐	☐	☐	☐
Eye irritation after 6 to 7 hours at the computer.	☐	☐	☐	☐
Back pain from non-stop computer use.	☐	☐	☐	☐

B Where does it hurt?

4 marks

Write the health issue of each numbered place.



1 .....

2 .....

3 .....

4 .....

C Choose the correct answer

5 marks

Underline your answer.

1. The word ergonomics comes from
   (a) Latin (b) Greek (c) Sinhala (d) English
2. Health issues arise from the continuous use of a computer for over
   (a) one hour (b) two hours (c) four hours (d) ten minutes
3. The pain of carpal tunnel syndrome is felt in the
   (a) eyes (b) back (c) knees (d) fingers
4. Which of these is a symptom of Computer Vision Syndrome?
   (a) dry eyes (b) a weak password
   (c) a slow network (d) a full mail box
5. RSI stands for
   (a) Random Screen Injury (b) Repeated System Issue
   (c) Repetitive Stress Injury (d) Real Sound Input

REVISION

Unit 6 in One Page

Read this the night before the test

Topic	The key facts
Films	3D technology (3D spectacles), holographic image processing (horror scenes), cartoon films (the 3D feature).
Music and games	Digital audio (recording, rhythm, mixing; CDs, home theatre); digital games (3D, 4D); simulation games (training).
Problems of ICT	Legal, values, physical, harmful software, environmental, health related and social issues (Figure 6.8).
Legal issues	Data theft (PII), unauthorized access, stolen intellectual property (a patent protects it), fraud.
Plagiarism	Stealing another's creative work and displaying it as one's own. Avoid it: citing, quoting, referencing.
Physical security	UPS, hardware firewall (in routers), locked doors, CCTV, surge protector (240 V), no dirt, dust or moisture.
Logical security	Passwords (letters, symbols, numbers), software firewall (always active), backups (in another place).
Malware	Virus, worm, Trojan horse, adware, bots, browser hijacker, phishing, spam.
Safeguards	Updated virus guard, check USBs, authorized software, user account, secure websites, no suspicious links.
Cyber security	Institution for Information Security of Sri Lanka (ICTA); ISMS certification.
Ergonomics	Greek "ergon" (job, action) + "nomos" (law): the job must suit the comfort of the worker.
Health issues	Musculoskeletal problems, RSI, CTS, CVS: over four hours of continuous use, wrong posture.

TICK WHEN YOU CAN DO IT

- [ ] I can explain how ICT is used in films, music and games.
- [ ] I can name the issues associated with the use of ICT.
- [ ] I can explain data theft, unauthorized access, intellectual property and fraud.
- [ ] I can avoid plagiarism by citing, quoting and referencing.
- [ ] I can name physical and logical security measures.
- [ ] I can name harmful software and say how to safeguard a computer.
- [ ] I can explain ergonomics and four health issues of computer use.

"Read every question twice. Answer once, clearly."

- THE KING OF ICT

WORD POWER

Unit 6 Word List

Thirty-one words you must know

Word	What it means
3D	Three dimensional: films look real (3D spectacles).
Holographic image processing	Displays a scene recorded in one place somewhere else.
Digital audio	Songs recorded, rhythm created and mixed with software.
Simulation game	A game in a natural setting created artificially.
Data theft	Stealing Personally Identifiable Information (PII).
PII	Personally Identifiable Information (names, photos ...).
Unauthorized access	Entering a computer system with a stolen password.
Intellectual property	Creations of the mind: an innovation or a new product.
Patent	A license issued by the state for innovations.
Fraud	Misusing personal documents; cheating; credit card fraud.
Plagiarism	Displaying another's creative work as one's own.
Citing	Mentioning the rightful owner and his information.
Quoting	Using inverted commas ("...") for a borrowed section.
Referencing	Listing the resources of the information at the end.
UPS	Uninterrupted Power Supply: its battery gives power.
Surge protector	Protects devices when the supply exceeds 240 V.
Firewall	Protects a network from unauthorized access.
CCTV	Cameras that monitor and record a computer laboratory.
Backup	A copy of essential data, kept in another place.
Malware	Malicious software that harms computers and networks.
Computer virus	Spreads within a computer as an executable file.
Computer worm	Spreads alone through e-mail and instant messages.
Trojan horse	Looks harmless; does not spread by itself.
Phishing	Deceiving users by e-mail to steal account details.
Spam	Unauthorized e-mail, mostly advertisements.
Cyber security	Securing Internet access, networks and operating systems.
Ergonomics	A job made to the comfort of the worker (Greek).
Posture	The way you sit; a wrong posture causes pain.
RSI	Repetitive Stress Injury: pain from shoulder to fingers.
CTS	Carpal Tunnel Syndrome: numbness and pain in the fingers.
CVS	Computer Vision Syndrome: irritation of the eyes.

"Learn the words well. A clear word wins a clear mark."

- THE KING OF ICT

END OF UNIT

Unit 6 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Cartoon films are popular among adults and the young because of
   (a) the 3D feature (b) high prices (c) old cameras (d) short stories
2. Recording songs, creating rhythm and mixing are done with
   (a) a patent (b) software (c) a firewall (d) a UPS
3. Which harmful software misdirects a user to a different website?
   (a) spam (b) bots (c) hijacker (d) adware
4. A surge protector acts when the voltage of the power supply exceeds
   (a) 24 V (b) 2.4 V (c) 2400 V (d) 240 V
5. Unauthorized e-mail, mostly about advertisements, is
   (a) spam (b) adware (c) bots (d) a Trojan horse
6. Ergonomics is made of the Greek words "ergon" and
   (a) "logos" (b) "nomos" (c) "graphos" (d) "metron"
7. Which of these is NOT one of the issues associated with the use of ICT in Figure 6.8?
   (a) legal issues (b) social issues (c) fast e-mail (d) health issues
8. A - A software firewall is part of the operating system. B - A Trojan horse spreads by itself. C - Adware displays unnecessary notices on the screen. Which are correct?
   (a) A and C only (b) A and B only (c) B and C only (d) A, B and C

2 Fill in the blanks

8 marks

One word in each blank.

1. Holographic images are mostly used in scenes depicting ..... .
2. Digital games in both 3D and ..... are the most popular today.
3. Mix letters, symbols and ..... to make a strong password.
4. A ..... is a license issued by the state for innovations.
5. Hardware firewalls are now built into broadband ..... .
6. A software firewall checks all ..... data for authenticity.
7. ..... deceives users by e-mail to collect bank account details.
8. Securing Internet access is also called ..... security.

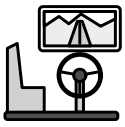
3 Name the pictures

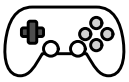
8 marks

Use the words below. Two words are not used.

Words: 3D spectacles, Simulation game, Digital game, Digital audio, UPS, Surge protector, CCTV camera, Firewall, Backup, Password

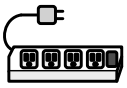
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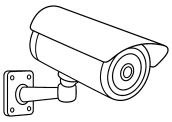
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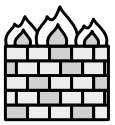
3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

Write in the space given.

1. What are simulation games? Write one use of them. (2 marks)

2. What is plagiarism? Write two ways to avoid it. (3 marks)

3. Write two legal issues in the use of ICT and explain one of them. (3 marks)

4. What is a backup? Write two places where a backup can be kept. (2 marks)

5. How is a computer worm different from a computer virus? (2 marks)

6. Write two ways to safeguard a computer from harmful software. (2 marks)

7. Name two health issues of long computer use. (2 marks)